AZAM HUSSAIN

Junior Machine Learning Engineer | NLP · LLMs · Retrieval-Augmented Generation

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PROFESSIONAL SUMMARY

Machine learning engineer specializing in Natural Language Processing (NLP), Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG). Has developed, deployed and optimized three end-to-end LLM applications spanning the full pipeline — document ingestion, chunking, embedding's, vector search and grounded response generation — using LangChain, ChromaDB and the OpenAI API, served through Flask and Streamlet, on a deep-learning foundation in TensorFlow and PyTorch. Currently an AI/ML Engineer Intern at FlyRank AI, completing an MSc in Computer Science and Microsoft Azure AI-900 certified, with AWS core services and Docker containerization in active development.

TECHNICAL SKILLS

NLP & LLMs: Natural Language Processing (NLP), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), transformer models, embedding's, semantic search, document chunking, prompt engineering, fine-tuning and transfer learning, conversational AI, LLM safety

Models & APIs: OpenAI API (GPT-4o, GPT-4o-mini, GPT-3.5-Turbo, Whisper, gpt-image-1), Google Gemini, Azure OpenAI, Hugging Face

Frameworks & Libraries: LangChain, TensorFlow, Keras, PyTorch, scikit-learn, NumPy, Pandas, Matplotlib

APIs & Web Frameworks: FastAPI, Flask, Streamlit, REST API design and integration, model serving endpoints

Databases & Vector Search: ChromaDB (vector database), embeddings storage and retrieval, SQL fundamentals; FAISS and Pinecone at concept level

Deployment & Version Control: Docker, Git, GitHub, environment and dependency management, debugging deployment and production issues, reproducible builds

Cloud: Microsoft Azure (AI-900 certified) — Azure OpenAI, Azure ML, Cognitive Services; AWS core services (S3, EC2, Lambda, IAM) currently in progress

ML Practice: End-to-end ML system integration, feature engineering, data pipelines, model training and evaluation, benchmarking, inference cost and latency optimization

PROFESSIONAL EXPERIENCE

AI/ML Engineer Intern — FlyRank AI 2026 – Present

- Took LLM-powered features from prototype to working product functionality, owning OpenAI API integration and prompt design end to end rather than handing off at the prototype stage, and collaborating cross-functionally on system integration.
- Replaced ad-hoc prompt tuning with a held-out test-query evaluation set, converting a subjective quality judgment into a measurable pass rate and making each prompt revision defensible against evidence rather than impression.
- Converted an ambiguous business brief into a defined ranking problem — specifying the target variable, feature set and evaluation metric — producing the framing document that the content opportunity scoring model now runs on.

AI/ML Intern — CareerCore, Lahore 2026 · 3 months

- Delivered three production-ready models inside a 3-month internship, each owned solo from raw data through preprocessing, training, fine-tuning and evaluation: a 7-class CNN skin-disease classifier at 89% accuracy, an NLP sentiment analyzer at 87% F1 across 50,000 reviews, and a fraud detector at 95% precision and 91% recall on 284,807 transactions.
- Gained 4–6% F1 on the sentiment task by benchmarking Logistic Regression, Naive Bayes and SVM rather than defaulting to one, and cut fraud false negatives by 22% against baseline by applying SMOTE to a dataset with 99.83% class imbalance.
- Rescued a stalled image classifier by switching from training-from-scratch to MobileNet transfer learning and fine-tuning, cutting over fitting by 18% and reaching 89% accuracy on only 10,000 labelled ceroscopy images. Tools: TensorFlow, Keras, OpenCV.

LLM & GENERATIVE AI PROJECTS

RAG-Based Knowledge Management System github.com/azam-hussain-ml/[repo]

- Designed and built a complete RAG pipeline over a private document corpus, returning grounded, source-cited answers: document ingestion → chunking → embedding generation → semantic retrieval from the ChromaDB vector database → LLM response synthesis.
- Tuned chunk size and overlap against a fixed question set rather than by inspection, measurably improving retrieval relevance and reducing ungrounded responses across successive iterations.
- Deployed as a Flask web application with API endpoints for query handling; resolved LangChain breaking-change migrations and SSL/environment failures to reach a stable build, and remediated a committed credentials file using git filter-repo — applying version-control and secrets-management discipline.
- Stack: LangChain, ChromaDB, OpenAI GPT-3.5-Turbo, Flask, Python.

VisionVoice AI — Multimodal Assistant github.com/azam-hussain-ml/[repo]

- Developed a multimodal assistant orchestrating four AI capabilities behind one interface: speech-to-text (Whisper), conversational chat (GPT-4o-mini), text-to-speech, and image generation.
- Diagnosed and fixed a Whisper transcription defect that returned Urdu/Hindi script for English audio by routing through the translations endpoint, and completed a DALL·E 3 → gpt-image-1 migration when the upstream model was retired — both without downtime to the demo build.
- Stack: OpenAI API (Whisper, GPT-4o-mini, gpt-image-1), Streamlet, Python.

Generative AI Job Recommender github.com/azam-hussain-ml/[repo]

- Built a recommender that ingests live job listings through the Apify API and uses an LLM to match and rank roles against a parsed CV.
- Migrated the LLM inference layer from OpenAI to Gemini after profiling cost and response latency per query, reducing per-query API spend while holding output quality steady on a manual review set.

EDUCATION, CERTIFICATIONS & ADDITIONAL

MSc Computer Science — The Superior University, Lahore | 2022]

Microsoft Azure AI Fundamentals (AI-900) — Microsoft, 2026 | Credential ID 9ED38B7EB43D0324

AI Fluency & Claude Code 101 — Anthropic Academy | AI / ML / Deep Learning Programme — NAVTTC, Pakistan.